Machine Learning Approaches for Predicting Athlete Injury Risk from Training Workload Data

Shourya Kukkala

Abstract

Athlete injury prevention remains a major challenge in sports science, as injuries often hinder training and can reduce athletic performance. In the age of athlete monitoring systems, huge volumes of training data can be collected and analyzed. In this study, a machine learning model is developed to predict athlete injury probability using training load data. The dataset picked has 42,798 observations and 72 training metrics describing worldload intensity, recovery patterns, and session frequency. Synthetic Minority Over-sampling Technique, otherwise known as SMOTE, was applied during model training to address potential class imbalance between injury and non-injury observations. A Random Forest classifier was trained and also evaluated using a test set and compared against a logistic regression baseline model. The Random Forest model had a ROC-AUC score of 0.745 outperforming the baseline model which achieved a score of 0.616. Using SHAP values, model interpretation included identifying strength training frequency, rest days, and interval training as key predictors of injury risk. These findings show the potential of machine learning models to support athlete monitoring and injury prevention strategies available to all.

1 Introduction

Athlete injuries pose a persistent challenge in sports performance and training management. Injuries can interrupt training progression, reduce performance during competitive events, and increase medical costs. Athletes are often exposed to high training loads and intense physical demands while trying to compete at a high level. Identifying factors contributing to injury risk has become an important and underlooked area of research in sports science.

In recent years, many advances in wearable technology and athlete monitoring systems have allowed coaches and sports scientists to collect detailed training data. These systems track many variables like training session frequency, distance covered, training intensity,

and recovery periods. These metrics show valuable insights into athlete workload, identifying patterns preceding injuries still remains an unsolved challenge due to the complex and nonlinear relationships among training variables.

Machine learning techniques offer a promising approach for analyzing and interpreting large training datasets and identifying useful patterns associated with injury risk. Machine learning models, unlike traditional statistical methods, can interpret nonlinear interactions between multiple variables and learn complex relationships directly from historical data. Recent studies have shown the potential of machine learning to support injury prediction, though many models still rely on small datasets and often lack interpretability.

This study investigates the use of machine learning to predict athlete injury risk using longitudinal training load data. A Random Forest classifier is trained on a dataset with 42,798 athlete-week observations and multiple metrics describing workload and recovery patterns. Synthetic Minority Over-sampling Technique, or SMOTE, is applied during model training to address class imbalance between injury and non-injury observations. The performance of the Random Forest Model is compared with a logistic regression baseline model and model interpretation techniques are used to show the most influential predictors of injury risk.

There are 3 primary contributions of this work. First, a machine learning model is developed to estimate injury risk using large-scale training load data. Second, the performance of the model is evaluated against a traditional statistical baseline. Third, feature importance analysis is performed to identify training variables that contribute strongly to injury risk predictions. These results show the potential of machine learning approaches to support injury prevention strategies using large scale data.

2 Dataset

The dataset used in this study contains longitudinal athlete training records organized at the athlete - week level. Each observation corresponds to one week of recorded training activity for an individual athlete and includes a binary label showing if an injury occurred during that specific training week.

The dataset contains a total of **42,798** athlete-week observations and 72 training-related variables. The metrics described are athlete workload, training intensity, and recovery patterns. These variables show the different components of training load that might contribute to injury risk.

Some features in the dataset include variables describing:

- Number of weekly training sessions
- Number of rest days
- Strength training sessions
- Interval training sessions
- Distance covered in different training zones
- High intensity training load
- Maximum distance covered in a single training session

Many variables also describe historical workload patterns by including measurements collected from previous weeks of training. These features allow the model to capture and interpret trends in training load over time, which may influence injury probability.

The target metric in the dataset is a binary injury indicator. A value of 1 shows that an injury occurred during the corresponding week and a value of 0 shows that no injury occurred. Because injury events are rare compared to non-injury observations, the dataset shows a significant class imbalance.

The dataset was used to train and evaluate machine learning models designed to estimate the probability of injury based on athlete training load patterns.

3 Methods

3.1 Data Preprocessing

Before training the machine learning models, the dataset was preprocessed to show consistent feature scrolling and compatibility with the modeling pipeline used. All training variables were numerical input features, while the injury indicator variable was also used as the binary target label.

Because the variables in the dataset were measured on different scales, feature standardization was applied using a standard scaling transformation. This process rescales each feature and has zero mean and unit variance improving model stability and training performance.

The dataset was then divided into training and testing subsets using an 80-20 split. The training subset was used for model development, while the testing subset was used for assessing model performance on unseen data.

3.2 Handling Class Imbalance

The dataset shows a significant class imbalance because injury events happen less frequently than non-injury observations. Machine learning models trained on imbalance datasets become biased towards predicting the majority class and this results in poor detection of injury events.

To fix this problem, SMOTE was applied to the training data. SMOTE generates synthetic samples of the minority class by interpolating between already existing observations helping balance the class distribution.

After applying SMOTE, the dataset had equal numbers of injury and non-injury observations. This lets the model learn patterns associated with injury events with more accuracy.

3.3 Machine Learning Models

A Random Forest classifier was used as the main predictive model in this study. Random Forest models are learning models that combine decision trees to improve predictive accuracy and reduce overfitting. Each tree is trained on a random subset of data and features, and the final prediction is calculated by aggregating the predictions of all the individual trees.

The Random Forest model was configured with 300 decision trees and a maximum tree depth of 12. Class weighting was also applied to further mitigate the effects of class imbalance.

To also show performance with a measurable benchmark, a logistic regression model was trained as a baseline classifier. Logistic regression is a commonly used statistical model for binary classification and is a useful comparison for evaluating the true performance of more complex machine learning models.

3.4 Evaluation Models

Model performance was evaluated using many standard classification metrics.

The primary evaluation metric was the Receiver Operating Characteristic Area Under the Curve (ROC - AUC), which measures the model's ability to distinguish between injury and non-injury observations across many different classification thresholds.

More evaluation metrics included precision, recall, and the F1-score. Recall is crucial because it shows the model's ability to correctly identify injury events.

A confusion matrix was also implemented to visualize the distribution of true positives, false positives, true negatives, and false negatives.

3.5 Model Interpretation

To better understand how different training variables influenced model predictions, SHAP, or Shapely Additive Explanations values were computed for the Random Forest model. SHAP values show a measure of the contribution of each feature to the predicted outcome.

By analyzing SHAP feature importance values, the variables most strongly associated with injury risk can be clearly distinguished. This provides invaluable insight into training load factors that can contribute most significantly to the model's predictions.

4 Results

4.1 Model Performance

The Random Forest classifier was evaluated using a held out test dataset containing 20 percent of the total observations. Model performance was assessed using many classification metrics, such as ROC-AUC, precision, recall, and F1-score.

The Random Forest model achieved a **ROC-AUC score of 0.745** showing a moderate ability to distinguish between injury and non-injury observations. In comparison, the logistic regressing baseline model achieved a **ROC-AUC score of 0.616,** showing that the Random Forest model showed substantially better predictive performance.

Because injury events happen infrequently in the dataset, the classification threshold had to be adjusted to improve detection of injury cases. The optimal threshold determined using the precision-recall curve was 0.296.

At this threshold, the model achieved these performance metrics:
Precision: 0.034
Recall: 0.327
Accuracy: 0.88

Although precision remains relatively low, the model is able to detect approximately one third of injury occurrences in the dataset.

4.2 Confusion Matrix

The confusion matrix shows a detailed view of classification performance.

True negatives: 7465
False positives: 988
False negatives: 73
True positives: 34

These results show the model identifies a portion of injury events successfully while keeping the missed injury count low. Figure 1 shows the confusion matrix for the Random forest model.

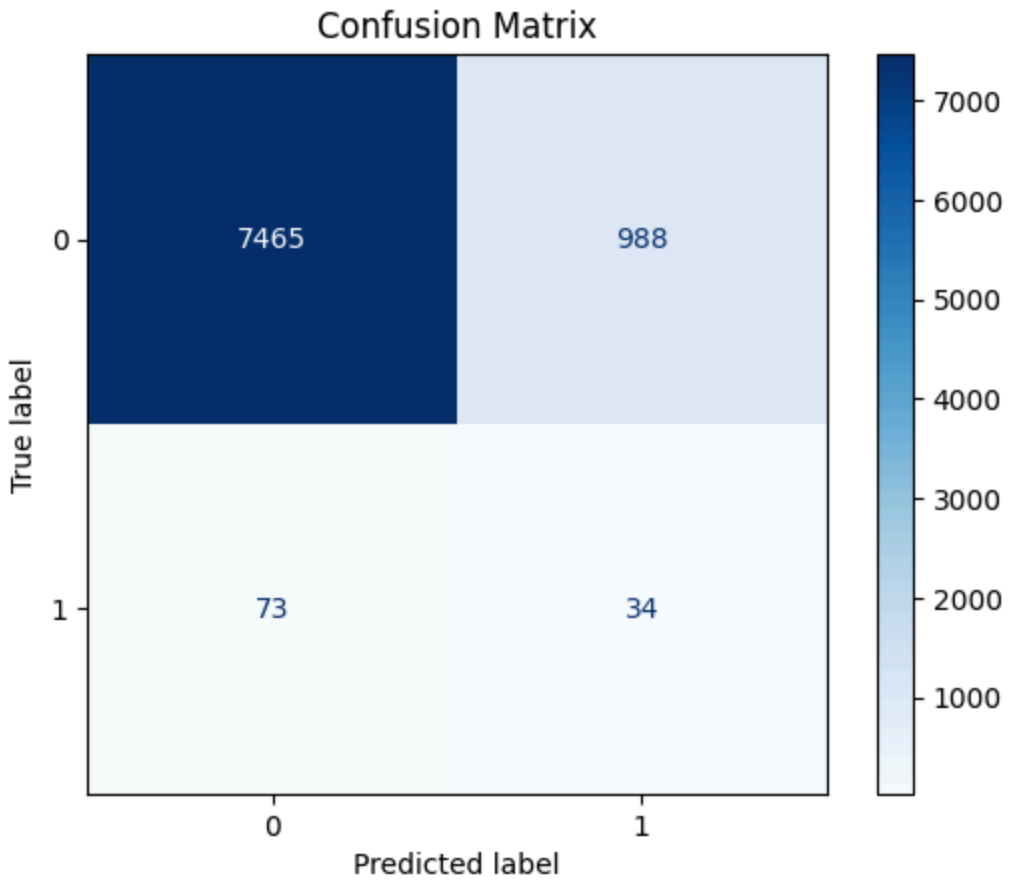

Figure 1. Confusion matrix for the Random Forest injury prediction model

4.3 Receiver Operating Characteristic Curve

The Receiver Operating Characteristic, or ROC, curve shows the tradeoff between the true positive rate and false positive rate across different classification thresholds.

Figure 2 presents the ROC curve for the Random FOrest mode. The area under the curve value of 0.745 shows that the model has moderate discriminatory power when distinguishing injury weeks from non-injury weeks.

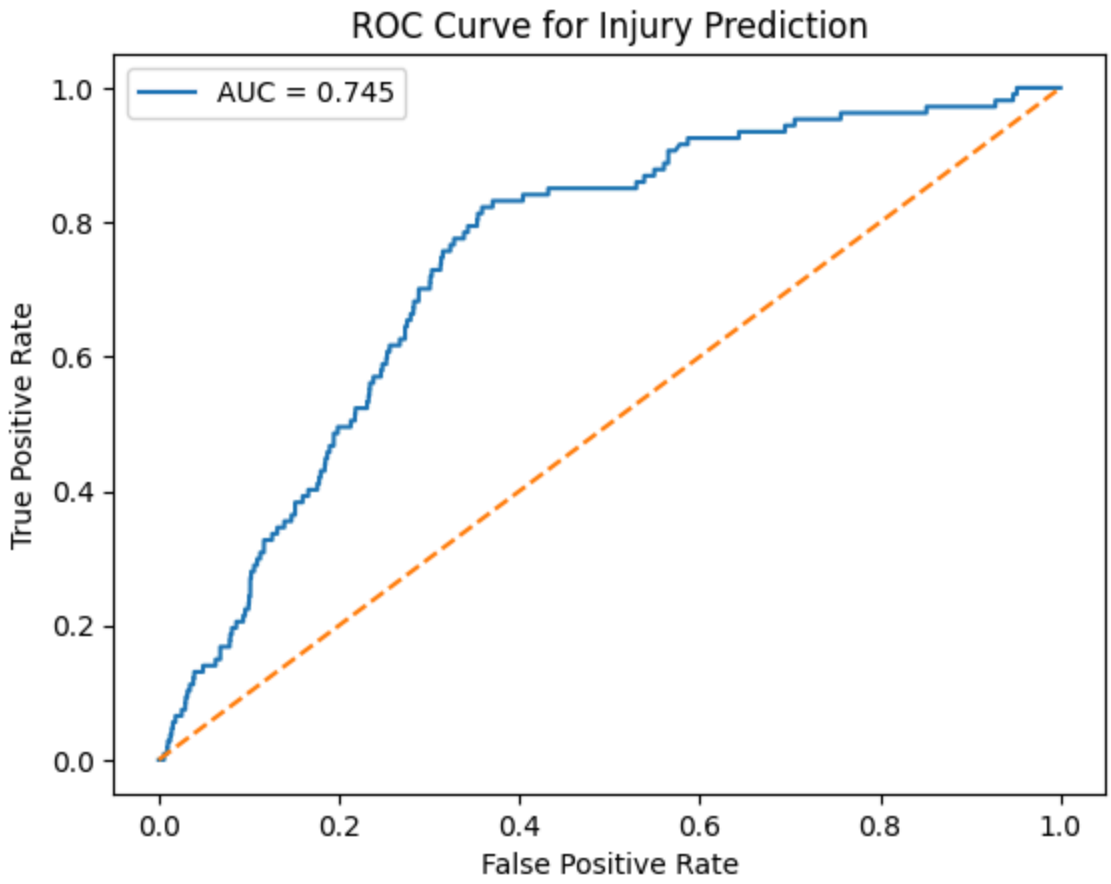

Figure 2. ROC curve showing classification performance of the Random Forest model

4.4 Feature Importance Analysis

To understand which training variables contributed most strongly to injury predictions, SHAP feature importance values were computed for the Random Forest model.

The SHAP analysis identified several variables that consistently influenced injury predictions. The most influential predictors included are:

Strength training frequency
Rest days
Interval training sessions
High-intensity training load
Weekly training sessions

These results suggest that both training intensity and recovery patterns play an important role in determining injury risk.

Figure 3 shows the SHAP feature importance plot showing the relative contribution of different variables to the model's predictions.

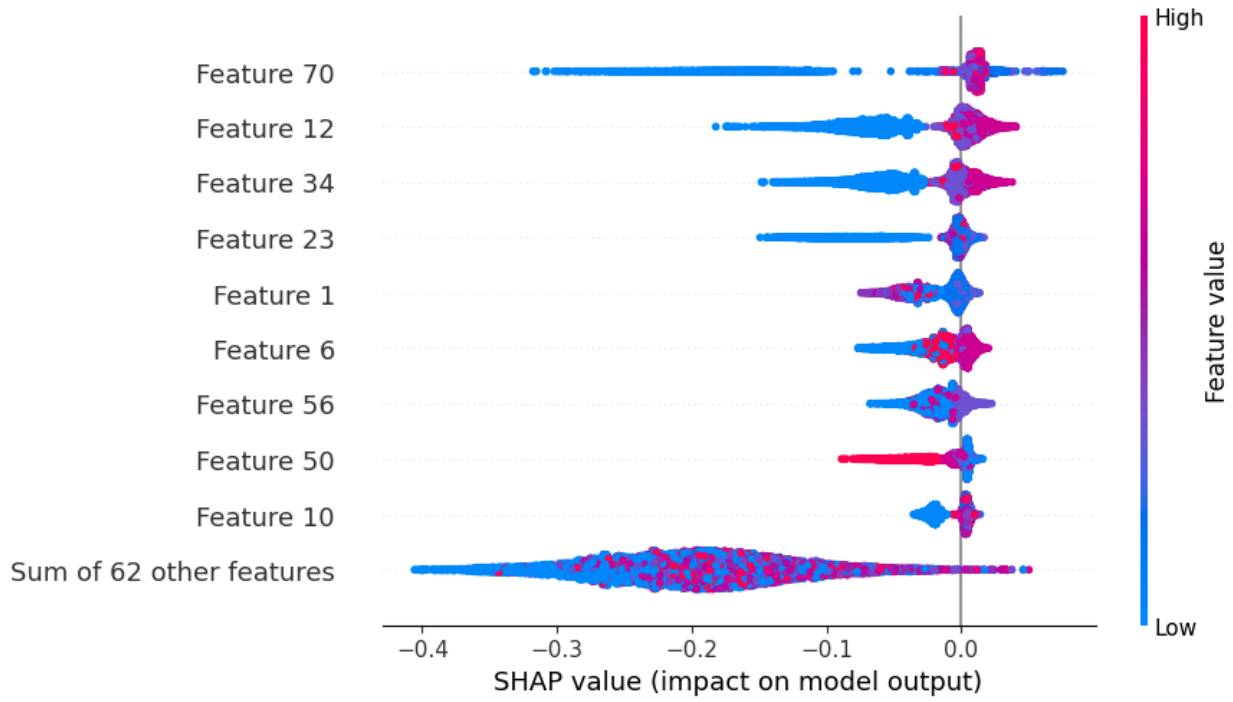

Figure 3. SHAP summary plot showing the most influential predictors of injury risk

4.4 Feature Importance Analysis

The results demonstrate the machine learning models can identify patterns in athlete training data associated with injury risk. The Random Forest model significantly outperformed the logistic regression baseline and achieved moderate predictive performance despite the highly imbalanced nature of the dataset.

Feature importance analysis further suggests that variables related to **training load intensity and recovery periods** are strongly associated with injury probability.

5 Discussion

The results of this study demonstrates that machine learning models can identify meaningful patterns in athlete training data that are also associated with injury risk. The Random Forest classifier achieved a ROC-AUC score 0.745, significantly outperforming the logistic regression baseline. This suggests that nonlinear relationships among training variables play an important role in predicting injury outcomes.

One of the key findings of this study is the importance of training load intensity and recovery variables in determining injury risk. Feature importance analysis using SHAP values revealed that variables such as strength training frequency, interval training sessions, and high-intensity training load consistently contributed to higher predicted injury risk. These findings align with existing sports science literature, which shows that excessive training intensity or rapid increases in workload may increase the likelihood of injury.

Recovery-related variables, including the number of rest days, also emerged as important predictors. This indicates that insufficient recovery may contribute to injury risk, supporting the idea that training load must be balanced with adequate recovery to prevent overuse injuries.

Although the model achieved moderate predictive performance, the relatively low precision reflects the inherent difficulty of predicting rare events such as injuries. Many athlete-weeks with elevated training load do not result in injury, which limits the model's ability to make highly precise predictions. However, the model's recall indicates that it can identify a meaningful proportion of injury events, making it useful as a risk detection tool rather than a definitive predictor.

Overall, the results suggest that machine learning models can provide valuable insights into injury risk patterns and may support data-driven decision-making in athlete monitoring systems.

6 Limitations

Several limitations should be considered when interpreting the findings of this study.

First, the dataset consists of observational training data and may not capture all factors that influence injury risk. Variables such as sleep quality, nutrition, biomechanics, and psychological stress are not included in the dataset but may play a significant role in injury occurrence.

Second, the model predicts injury probability based on historical patterns but does not establish causal relationships between training variables and injuries. As a result, the findings should be interpreted as associations rather than direct causes.

Third, the dataset exhibits a high degree of class imbalance, with injury events representing a small fraction of total observations. Although SMOTE was used to address this imbalance, synthetic data generation may introduce additional variability.

Finally, the model was evaluated on a single dataset. Additional validation using independent datasets would be necessary to assess generalizability and real-world applicability.

7 Conclusion

This study demonstrates that machine learning techniques can be used to predict athlete injury risk using training load data. A Random Forest classifier achieved a ROC-AUC score of 0.745 and significantly outperformed a logistic regression baseline model.

Feature importance analysis identified training intensity, session frequency, and recovery patterns as key predictors of injury risk. These findings suggest that machine learning models can capture meaningful relationships within training data and may support injury prevention strategies.

Although predictive performance remains moderate, the model provides a framework for identifying periods of elevated injury risk. Future work may incorporate additional physiological and behavioral variables to improve predictive accuracy and support more comprehensive athlete monitoring systems.

8 Code Availability

All code used for data preprocessing, model training, and evaluation is available at:

https://github.com/yourusername/injury-prediction-ml

The repository contains scripts necessary to reproduce all experiments presented in this study.

9 References

Breiman, L. Random Forests. Machine Learning, 2001

Chawla, N. V. et al. SMOTE: Synthetic Minority Over-sampling Technique. Journal of Artificial Intelligence Research, 2002

Lundberg, S. and Lee, S. A Unified Approach to Interpreting Model Predictions. NeurIPS, 2017